

Cluster Robust Variance Estimation

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Introduction

- ▶ To conduct hypothesis tests we need the asymptotic distribution of our estimators
- ▶ Usal variance estimator is not consistent in case of clustered data
- ▶ Cluster robust estimator exist but is not actually robust
- ▶ When to use it?

Which is the problem?

- ▶ Inside a cluster the scores of 2 individuals are not independent
- ▶ Usual White Variance estimator is not consistent

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i \xrightarrow{d} \mathcal{N}(0, \Omega_N)$$

$$\Omega_N = \underbrace{\frac{1}{N} \sum_{i=1}^N \text{Var}(X_i u_i)}_{\text{diagonal terms}} + \underbrace{\frac{2}{N} \sum_{i=1}^N \sum_{j \neq i}^N \text{Cov}(X_i u_i, X_j u_j)}_{\text{off-diagonal terms}}$$

Literature Review

- ▶ Cameron and Miller (2015); Wooldridge (2003):
Clustering corrects OLS standard errors for within-cluster dependence; validity relies on **large- G asymptotics**.
- ▶ **Ibragimov and Müller (2010):**
With fixed or small G , inference based on cluster-level estimates and a t_{G-1} test remains valid under heteroskedasticity.
- ▶ **Abadie et al. (2023):**
Clustering is justified by **sampling or assignment design**, not residual correlation per se.
- ▶ **Key implication:**
If neither sampling nor treatment is clustered, **do not cluster**; CRVE is often **conservative**, especially with heterogeneous effects.

Representation of Clustered Data

- ▶ Sample with N individuals grouped in G clusters:
 $X_{ig} = (X_{11}, \dots, X_{N_1 1}, X_{12}, \dots, X_{N_2 2}, \dots, X_{N_G G})$
- ▶ For i, j in the same cluster g :

$$P(X_i > E[X] \mid X_j > E[X]) > P(X_i > E[X])$$

- ▶ Cluster g can be seen as a **subpopulation**:

$$X_g = \{X_i : i \in g\}$$

Clustered Correlation from Heterogeneous Effects

- ▶ True model:

$$Y_{ig} = X_{ig}\beta_g + u_i$$

- ▶ Estimated model:

$$Y_{ig} = X_{ig}\beta_{\text{APE}} + e_{ig}$$

- ▶ Residual:

$$e_{ig} = X_{ig}(\beta_g - \beta_{\text{APE}}) + u_i$$

- ▶ Hence, **heterogeneous β_g generates clustered residuals.**

Assumptions on the DGP

- ▶ **Two-stage sampling in all frameworks:**
 1. sample clusters
 2. sample individuals within clusters
- ▶ **Fixed number of clusters (G fixed)**
 - ▶ clusters are drawn once
 - ▶ repeated samples come only from re-sampling individuals
 - ▶ **only unit-level randomness**
- ▶ **Increasing number of clusters ($G \rightarrow \infty$)**
 - ▶ clusters and individuals are re-sampled each time
 - ▶ **two sources of randomness:**
 - ▶ cluster selection
 - ▶ unit-level sampling

Variance of Cluster-Level Effects

- ▶ Assume a population of cluster-specific effects:

$$\beta_g \sim \text{i.i.d. with } E[\beta_g] = \beta_{\text{APE}}, \quad \text{Var}(\beta_g) = \sigma_\beta^2.$$

- ▶ We draw G coefficients:

$$\beta = (\beta_1, \dots, \beta_G)'.$$

- ▶ Then

$$E[\beta] = \begin{bmatrix} \beta_{\text{APE}} \\ \vdots \\ \beta_{\text{APE}} \end{bmatrix}, \quad \text{Cov}(\beta) = \sigma_\beta^2 I_G.$$

Asymptotics in the Number of Clusters

- ▶ Sample mean of the cluster effects:

$$\bar{\beta} = \frac{1}{G} \sum_{g=1}^G \beta_g \xrightarrow{p} \beta_{\text{APE}}$$

- ▶ Central Limit Theorem:

$$\frac{1}{\sqrt{G}} \sum_{g=1}^G (\beta_g - \beta_{\text{APE}}) \xrightarrow{d} \mathcal{N}(0, \sigma_{\beta}^2).$$

- ▶ If all β_g were observed, we could estimate σ_{β}^2 via:

$$\hat{\sigma}_{\beta}^2 = \frac{G}{G-1} \sum_{g=1}^G (\beta_g - \bar{\beta})^2.$$

From True β_g to Estimated $\hat{\beta}_g$

- ▶ In practice, β_g is not observed; we estimate it from data.
- ▶ For each cluster g :

$$\sqrt{N_g}(\hat{\beta}_g - \beta_g) \xrightarrow{d} \mathcal{N}(0, V_g),$$

where

$$V_g = (X_g' X_g)^{-1} E_g[X_g' u_g u_g' X_g] (X_g' X_g)^{-1}.$$

- ▶ Across clusters we thus have G asymptotically normal estimators,
each centered at its own β_g with cluster-specific variance V_g .

Interpretation: Signal + Noise

- Write each estimator as

$$\hat{\beta}_g = \beta_g + e_g,$$

with

$$\sqrt{N_g} e_g \xrightarrow{d} \mathcal{N}(0, V_g), \quad E[e_g] = 0.$$

- The vector $\hat{\beta}$ can be viewed as **noisy observations** of the underlying cluster parameters, with noise coming from within-cluster sampling.

Decomposition Around β_{APE}

- Model the true effects as

$$\beta_g = \beta_{\text{APE}} + d_g, \quad d_g \sim \text{i.i.d. } \mathcal{N}(0, \sigma_\beta^2).$$

- Then

$$\hat{\beta}_g = \beta_{\text{APE}} + d_g + e_g.$$

- The cluster mean equals the pooled OLS estimator:

$$\hat{\beta}_{\text{pols}} = \frac{1}{G} \sum_{g=1}^G \hat{\beta}_g = \beta_{\text{APE}} + \frac{1}{G} \sum_{g=1}^G d_g + \frac{1}{G} \sum_{g=1}^G e_g.$$

- As $G \rightarrow \infty$, both averages of d_g and e_g converge to zero in probability.

Monte Carlo Decomposition of $\text{Var}(\hat{\beta}_{\text{ols}})$

- ▶ Goal: assess how well

$$\text{Var}(\hat{\beta}_{\text{ols}}) \approx \text{Var}(d_g) + \mathbb{E}[\text{Var}(e_g)]$$

holds in finite samples.

- ▶ Two components:
 - ▶ $d_g = \beta_g - \beta_{\text{APE}}$ (heterogeneity)
 - ▶ $e_g = \hat{\beta}_g - \beta_g$ (sampling error)

Simulation Design

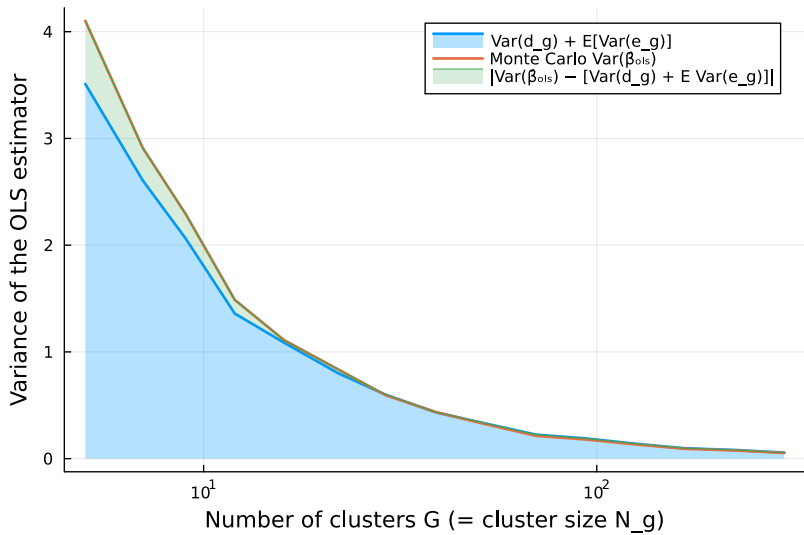
- ▶ Loop over different **numbers of clusters** G and **cluster sizes** N_g .
- ▶ For each (G, N_g) , run **1000 Monte Carlo replications**.
- ▶ For each replication:
 1. Draw heterogeneous coefficients β_g .
 2. Generate a population with G clusters.
 3. Sample N_g individuals from each cluster.
 4. Estimate all $\hat{\beta}_g$ and pooled $\hat{\beta}_{ols}$.

Recorded Quantities

- ▶ For every repetition and design (N_g, G) :
 - ▶ $e_g = \hat{\beta}_g - \beta_g$ (estimation errors)
 - ▶ $d_g = \beta_g - \beta_{\text{APE}}$ (heterogeneity terms)
 - ▶ $\hat{\beta}_{\text{ols}}$ (pooled estimator)
 - ▶ White and CRVE variance estimates of $\hat{\beta}_{\text{ols}}$.
- ▶ For each (N_g, G) we compute:
 - ▶ $\text{Var}(\hat{\beta}_{\text{ols}})$ (empirical variance)
 - ▶ $\text{Var}(d_g)$ and $\mathbb{E}[\text{Var}(e_g)]$
 - ▶ relative discrepancy.

Performance of the Decomposition

- ▶ When either G or N_g is large:
 - ▶ d_g and e_g are measured very precisely.
 - ▶ $\text{Var}(d_g) + \mathbb{E}[\text{Var}(e_g)]$
closely matches $\text{Var}(\hat{\beta}_{\text{ols}})$.
- ▶ When both G and N_g are small:
 - ▶ cluster-level estimates are noisy;
 - ▶ between-cluster dispersion is overstated;
 - ▶ the decomposition is a poorer approximation.



CRVE–White Differences as Heterogeneity Measure

- ▶ We test whether the **difference** between CRVE and White variance estimators captures the heterogeneity component

$$\text{Var}(\bar{d}_g), \quad d_g = \beta_g - \beta_{\text{APE}}.$$

- ▶ For each design (N_g, G) :
 - ▶ compute average CRVE variance,
 - ▶ compute average White variance,
 - ▶ take their difference,
 - ▶ compare it to $\text{Var}(\bar{d}_g)$ from the simulations.

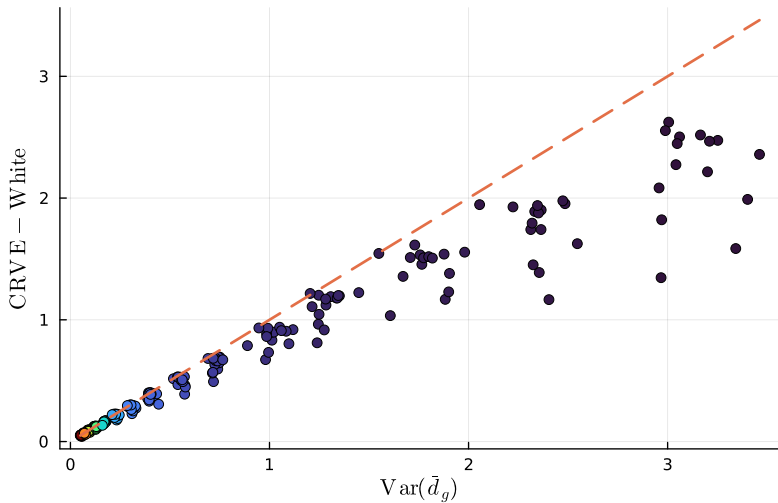
Simulation-Based Comparison

- ▶ For every (N_g, G) :
 - ▶ horizontal axis: $\text{Var}(\bar{d}_g)$ (variance of mean deviations),
 - ▶ vertical axis:

$$\mathbb{E}[\widehat{\text{Var}}_{\text{CRVE}}] - \mathbb{E}[\widehat{\text{Var}}_{\text{White}}].$$

- ▶ Each point is a design (N_g, G) :
 - ▶ color intensity encodes G (lighter = larger G).
- ▶ The 45-degree dashed line is the **benchmark** of perfect equality.

CRVE – White vs $\text{Var}(\text{mean}(\bar{d}_g))$



Results and Interpretation

- ▶ Points lie very close to the 45-degree line:
 - ▶ CRVE–White differences almost exactly recover $\text{Var}(\bar{d}_g)$.
- ▶ Overall, the simulations support:

$$\mathbb{E}[\widehat{\text{Var}}_{\text{CRVE}}] - \mathbb{E}[\widehat{\text{Var}}_{\text{White}}] \approx \text{Var}(\bar{d}_g),$$

and this relationship strengthens as the number of clusters G grows.

- Abadie, Alberto, Susan Athey, Guido W. Imbens, and Jeffrey M. Wooldridge. 2023. "When Should You Adjust Standard Errors for Clustering?" *The Quarterly Journal of Economics* 138 (1): 1–35. <https://doi.org/10.1093/qje/qjac038>.
- Cameron, A. Colin, and Douglas L. Miller. 2015. "A Practitioner's Guide to Cluster-Robust Inference." *Journal of Human Resources* 50 (2): 317–72. <https://doi.org/10.3368/jhr.50.2.317>.
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